



Further analysis of Rice's whale ship-strike risk

40 messages

Ben Best <ben@ecoquants.com>

Mon, Feb 17, 2025 at 7:40 PM

To: Rachel Rhodes <rachelrhodes@ucsb.edu>

Cc: "Jasny, Michael" <mjasny@nrdc.org>, "Chen, Leyi" <lchen@nrdc.org>

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

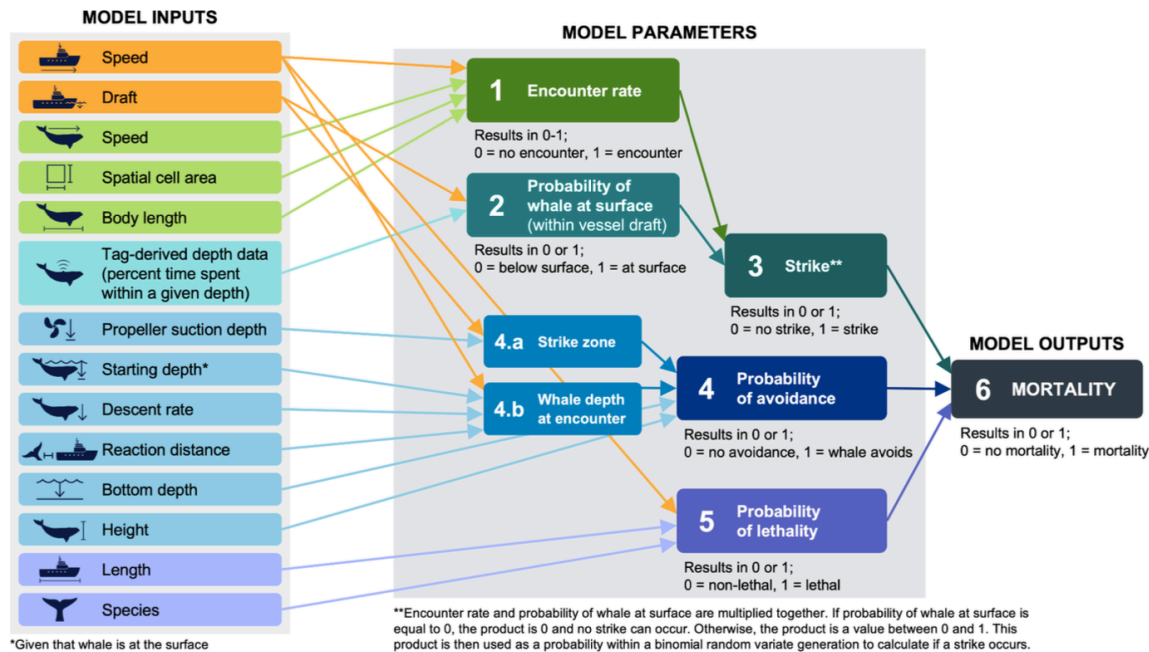


Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
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over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
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- TUI Cruises (only one ship was in the area)
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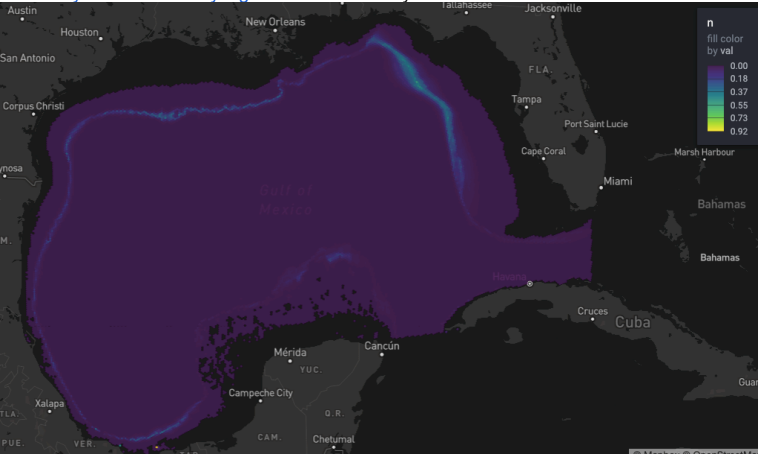
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...
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date    cell_ll_lat cell_ll_lon mmsi      hours  dist_travelled_km avg_speed_knots
2023-01-01    -41    173.2 512005516 0.083    0.823    5.461
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As for "Tag-derived depth data", I only know of Kok et al (2023) with strong diel patterns:

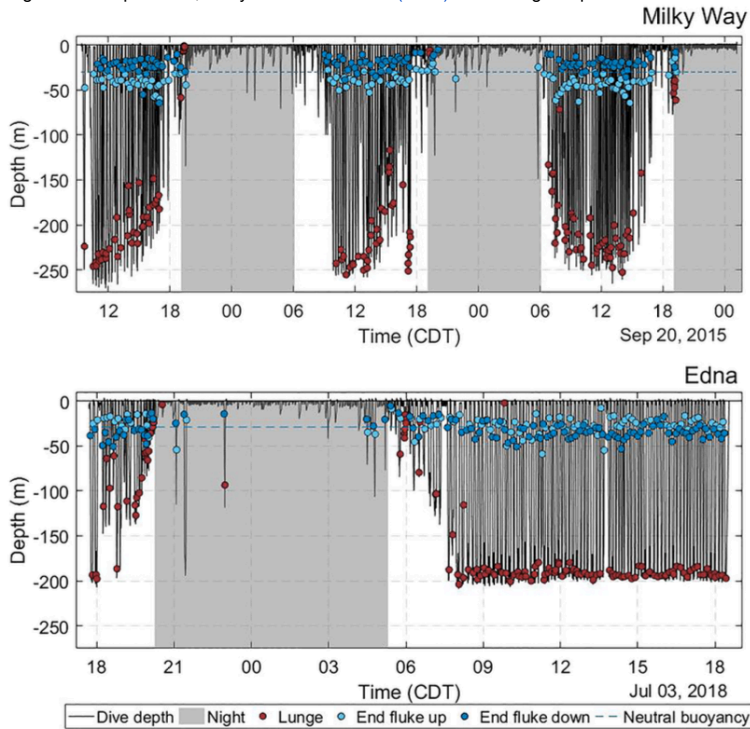


Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). 21Jan25 with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**

Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*

2. **shipstrike**

Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*

3. **whalestrike**

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)

Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- **whalestrike**
 Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,

Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

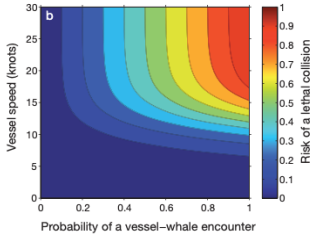
Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
[WhaleSafe End Goal: Effect Behavioral Change](#)



1. Social Incentives show best performer.

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Blue whale mortality in California



Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

4479 W. 5TH AVENUE

VANCOUVER, BC V6R1S4

C 310.560.5536

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Please save paper.
Think before printing.

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To: Ben Best <ben@ecoquants.com>, Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: "Chen, Leyi" <lchen@nrdc.org>

Wed, Mar 12, 2025 at 12:46 PM

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

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To: Rachel Rhodes <rachelrhodes@ucsb.edu>
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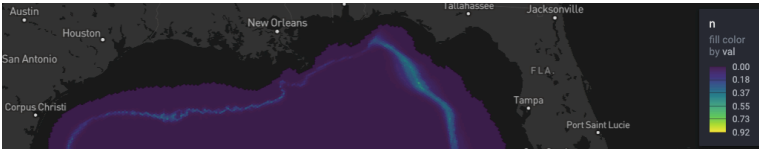
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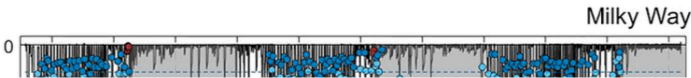


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If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
 Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
 Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
 Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
 Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
 Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

· [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the *units* of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a't First Nation territory](#). *ESR*
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

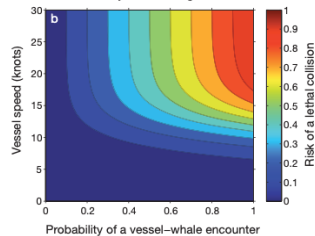
Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
[WhaleSafe End Goal: Effect Behavioral Change](#)



- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Blue whale mortality in California



Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL
4479 W. 5TH AVENUE
VANCOUVER, BC V6R1S4
C 310.560.5536
MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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2 attachments



Soldevilla et al. 2017 (RiWa and vessel and fisheries interactions).pdf

1594K



Individual_Vessel_Tables.xlsx

1863K

Ben Best <ben@ecoquants.com>

Wed, Mar 12, 2025 at 5:11 PM

To: "Jasny, Michael" <mjasny@nrdc.org>

Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>, "Chen, Leyi" <lchen@nrdc.org>

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at [la WhaleSafe.com](https://www.whalesafe.com), given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

[Quoted text hidden]

Jasny, Michael <mjasny@nrdc.org>

Thu, Mar 13, 2025 at 8:38 AM

To: Ben Best <ben@ecoquants.com>

Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>, "Chen, Leyi" <lchen@nrdc.org>

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,

Michael

From: Ben Best <ben@ecoquants.com>

Sent: Wednesday, March 12, 2025 4:11 PM

To: Jasny, Michael <mjasny@nrdc.org>

Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at [la WhaleSafe.com](https://www.whalesafe.com), given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
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I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:



Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- 1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

```
vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv
```

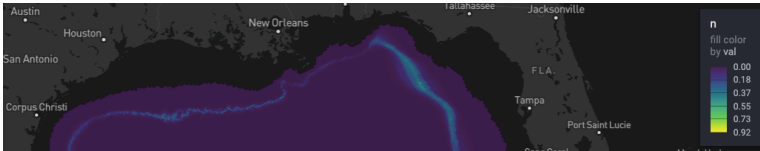
date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165

2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly **n** and **cv** values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

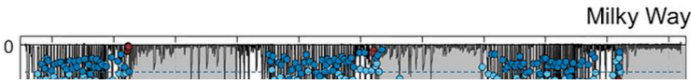


Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). 21Jan25 with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

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This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

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Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

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I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

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Sent: Thursday, January 23, 2025 2:52 PM
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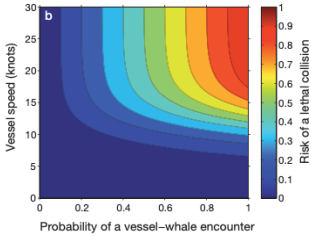
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Blue whale mortality in California



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Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

4479 W. 5TH AVENUE

VANCOUVER, BC V6R1S4

C 310.560.5536

MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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Consulting agreement. Ben Best. Mar25.docx
44K

Thu, Mar 13, 2025 at 8:58 AM

I will reach out to Tyler today to see if we can get the remaining AIS data. Double checking the date range for the analysis is the 2023 calendar year?

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Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:



Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- 1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUICruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

```
vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv
```

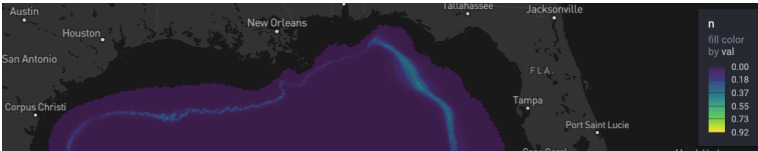
date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165

2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly **n** and **cv** values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

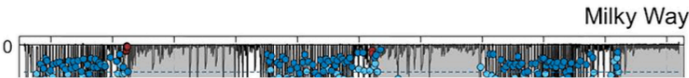


Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). **21Jan25** with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
 Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
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4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
 Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEFNU7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

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Cheers, Ben

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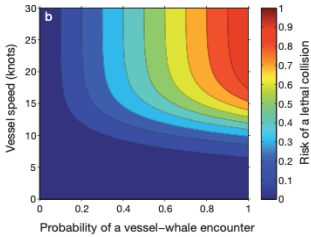
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Blue whale mortality in California



Cheers, Ben

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On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben’s schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

4479 W. 5TH AVENUE

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Please save paper.
Think before printing.

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Blue whale mortality in California



image005.png
364K

Jasny, Michael <mjasny@nrdc.org>
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Ben Best <ben@ecoquants.com>, "Chen, Leyi" <lchen@nrdc.org>

Tue, Mar 18, 2025 at 6:44 PM

Hi all – An update from my end. I spoke with NOAA about the dive data from last summer’s tagging effort. Unfortunately, the data haven’t been processed and probably won’t be available for quite some time, although it is a science priority for the agency. I asked whether the data are generally consistent with the depth usage reported in Kok et al. 2023 and was told that they are; so I think we should proceed with the approach we’ve discussed below.

Rachel, don’t know if you heard from Tyler about the monthly data?

Also, fyi, Regan and I spoke with Jess and Becca on Friday. They’re interested in collaborating on company outreach, which is great news. Either Regan or I will be down next month for the Blue Whales & Blue Skies awards event.

Best,
Michael

From: Jasny, Michael
Sent: Thursday, March 13, 2025 8:01 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

I think that's right—we want to use the same date range that you used in your analysis. Thanks, Rachel.

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, March 13, 2025 7:58 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

I will reach out to Tyler today to see if we can get the remaining AIS data. Double checking the date range for the analysis is the 2023 calendar year?

On Thu, Mar 13, 2025 at 7:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?

- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <chen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:



Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25?](#)

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

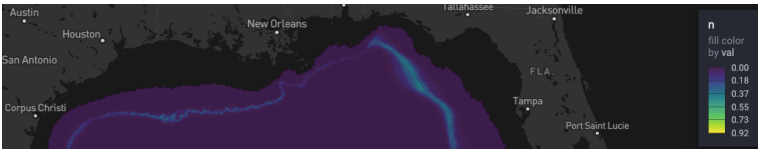
vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly **n** and **cv** values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

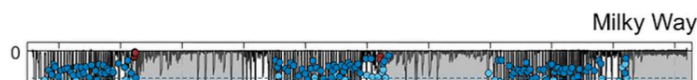


Figure 1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). [21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
 Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
 Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
 Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
 Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
 Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEFNU7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or

Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

• [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of 1;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model (Kelley et al, 2020) and a couple of R packages to facilitate calculation:

- [shipstrike](#)
 Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
 Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- [whalestrike](#)
 Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,

Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

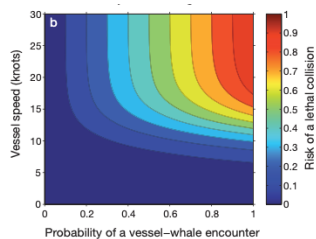
- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

WhaleSafe End Goal: Effect Behavioral Change



1. **Social Incentives**
show best performer.

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection.](#) *PLOS ONE*

Blue whale mortality in California



Figure 1. Map of California showing the distribution of blue whale mortality hotspots along the coast.



Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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WhaleSafe End Goal: Effect Behavioral Change

The diagram illustrates a feedback loop for the WhaleSafe program. It starts with 'Social Incentives' (a box with a whale icon) which leads to 'Immediate Reward' (a box with a whale icon). From 'Immediate Reward', the flow goes to 'Progress Monitoring' (a box with a whale icon). From 'Progress Monitoring', the flow goes back to 'Social Incentives'. In the center of this loop is a large box labeled 'WhaleSafe' with a whale icon. Below the 'WhaleSafe' box is a box labeled 'WhaleSafe' with a whale icon.

1. **Social Incentives** show best performer, community average and individual score
2. **Immediate Reward** show priority or other incentive package cash awards to best performer
3. **Progress Monitoring** shows performance over time, to improvement or decline

Blue whale mortality in California

A map of California shows the distribution of blue whale mortalities. A legend indicates that blue dots represent 'Blue whale mortality' and red dots represent 'Other species mortality'. A table below the map provides data for blue whale mortalities.

Species	FDR 2006-2016	Stranding Standings Representative	Model	Total
Blue whale	2.3	10	205.51	

Model results for blue whale mortality:

- (1) Decreasing with age
- (2) 10% avoidance
- (3) No avoidance

image005.png
364K

Thu, Mar 20, 2025 at 3:29 PM

Rachel

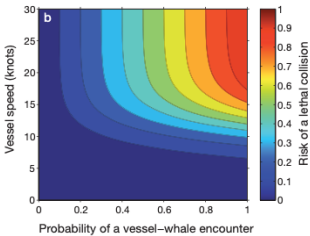
Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
WhaleSafe End Goal: Effect Behavioral Change



1. Social Incentives
show best performer.

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Blue whale mortality in California



Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

[Quoted text hidden]
[Quoted text hidden]

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Jasny, Michael <mjasny@nrdc.org>
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Ben Best <ben@ecoquants.com>, "Chen, Leyi" <lchen@nrdc.org>

Tue, Mar 25, 2025 at 11:06 AM

Thanks, Rachel! From our end, we'd like to move forward as quickly as possible, so the timing really depends on Ben's availability.

Ben, do you have a sense of when you can start? Also, please let me know if you have any edits to the contract and scope of work.

Thanks,
Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, March 20, 2025 2:29 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi all,

Thanks for these updates. I reached out to Tyler about AIS data. They are planning to make a similar dataset for the full time-series of our data that they can share publicly, which means they need more time to iron out the details of vessel classes and other features they are going to include. In the meantime, they are going to send the query they used so I can manually pull and generate the 2023 rasters that we need for this analysis.

They were going to discuss this week and get back to me with more details. Ideally, when would you like to have AIS data in hand?

That is great news that Jess & Becca were onboard for helping with outreach. I will be at the awards ceremony in Long Beach - it will be great to see you or Regan there!

Rachel

On Tue, Mar 18, 2025 at 5:45 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi all – An update from my end. I spoke with NOAA about the dive data from last summer's tagging effort. Unfortunately, the data haven't been processed and probably won't be available for quite some time, although it is a science priority for the agency. I asked whether the data are generally consistent with the depth usage reported in Kok et al. 2023 and was told that they are; so I think we should proceed with the approach we've discussed below.

Rachel, don't know if you heard from Tyler about the monthly data?

Also, fyi, Regan and I spoke with Jess and Becca on Friday. They're interested in collaborating on company outreach, which is great news. Either Regan or I will be down next month for the Blue Whales & Blue Skies awards event.

Best,
Michael

From: Jasny, Michael
Sent: Thursday, March 13, 2025 8:01 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

I think that's right—we want to use the same date range that you used in your analysis. Thanks, Rachel.

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, March 13, 2025 7:58 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

I will reach out to Tyler today to see if we can get the remaining AIS data. Double checking the date range for the analysis is the 2023 calendar year?

On Thu, Mar 13, 2025 at 7:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM

To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:



Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- 1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUICruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

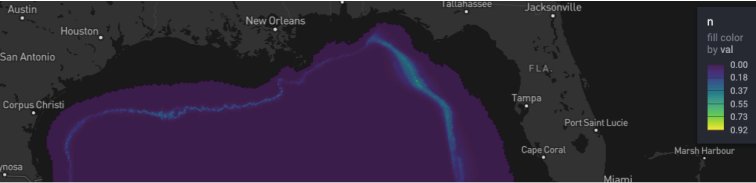
date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461

2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly n and cv values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

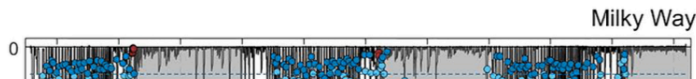


Figure 1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work. 21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*

2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*

3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison

or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

· [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- [shipstrike](#)
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a First Nation territory](#). *ESR*
- [whalestrike](#)
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you’ve both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We’ve suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

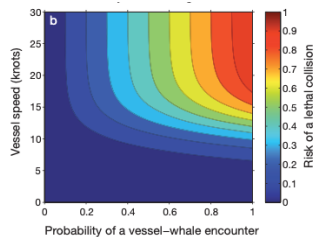
Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
[WhaleSafe End Goal: Effect Behavioral Change](#)



1. Social Incentives
show best performer.

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#).
Endangered Species Research



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection.](#) *PLOS ONE*

Blue whale mortality in California



Figure 1. Map of California showing the distribution of blue whale mortality hotspots along the coast.



Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!
Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL
4479 W. 5TH AVENUE
VANCOUVER, BC V6R1S4

C 310.560.5536
MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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2 attachments



Ben Best <ben@ecoquants.com>

To: "Jasny, Michael" <mjasny@nrdc.org>

Wed, Mar 26, 2025 at 4:58 PM

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

[illegible]

Thu, Mar 27, 2025 at 10:20 AM

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via DocuSign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at [WhaleSafe.com](https://www.whalesafe.com), given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of

this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185

2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly μ and cv values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). 21Jan25 with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <chen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
 Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
 Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
 Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
 Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
 Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEFNU7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- [whalestrike](#)
- Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
- Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast. Scientific Reports](#)
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates. Ocean & Coastal Management](#)
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory. ESR](#)
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk. Biological Conservation](#)

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

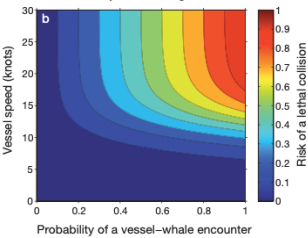
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM

To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

4479 W. 5TH AVENUE

VANCOUVER, BC V6R1S4

C 310.560.5536

MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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Ben Best, PhD

Environmental Data Scientist

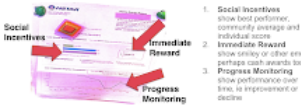
[EcoQuants.com](#)

busy times: [PT](#), [ET](#)

[+1.805.705.9770](#)

3 attachments

WhaleSafe End Goal: Effect Behavioral Change



1. Social Incentives
show best performer, community average and individual score

2. Immediate Reward
show penalty or other reward, perhaps cash awards too

3. Progress Monitoring
show performance over time, is improvement or decline

image005.png

514K

Blue whale mortality in California



$Mortality = \sum_{i=1}^n (W_i \times A_i) \times P_i$

Species	FDR 2006-2016	Stranding Standings (Repopulation: baseline)	Model	Total Mortality
Blue whale	2.5	10	250.51	10
			1) Decreasing with speed	10
			2) 75% avoidance	10
			3) No avoidance	40

image006.png

364K



44K

Fri, Mar 28, 2025 at 10:20 AM

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much. Ben

[Quoted text hidden]

Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

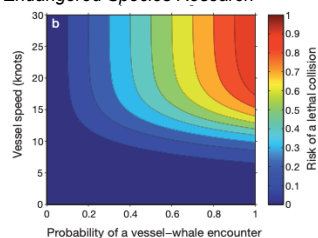
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- Vanderlaan et al (2008) *Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf. Endangered Species Research*



- Rockwood et al (2017) High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection. *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

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 **Consulting agreement. Ben Best. Mar25. rev.2 BDB-signed.docx**
58K

Jasny, Michael <mjasny@nrdc.org>
To: Ben Best <ben@ecoquants.com>

Mon, Mar 31, 2025 at 12:57 PM

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice’s whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per Blondin et al (2025) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 (Garrison et al, 2025) with code found at SEFSC/VesselStrikeRiskModel. I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly μ and σ values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work. 21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
 Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
 Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
 Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a First Nation territory](#). *ESR*
3. **whalestrike**
 Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
 Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNU7EiFD7E-ooLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- [whalestrike](#)
- Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
- Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast. Scientific Reports](#)
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates. Ocean & Coastal Management](#)
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory. ESR](#)
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk. Biological Conservation](#)

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

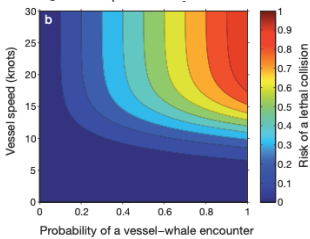
On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*
- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*



Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM

To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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VANCOUVER, BC V6R1S4

C 310.560.5536
MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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2 attachments

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Thanks,
Rachel

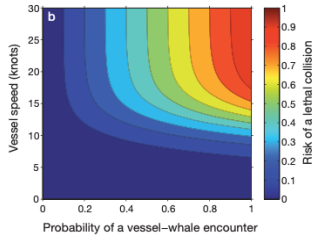
On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

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- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Blue whale mortality in California



Figure 1. Map of California showing the location of the study area for blue whale mortality.



Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!

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Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Thanks,
Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, March 20, 2025 2:29 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi all,

Thanks for these updates. I reached out to Tyler about AIS data. They are planning to make a similar dataset for the full time-series of our data that they can share publicly, which means they need more time to iron out the details of vessel classes and other features they are going to include. In the meantime, they are going to send the query they used so I can manually pull and generate the 2023 rasters that we need for this analysis.

They were going to discuss this week and get back to me with more details. Ideally, when would you like to have AIS data in hand?

That is great news that Jess & Becca were onboard for helping with outreach. I will be at the awards ceremony in Long Beach - it will be great to see you or Regan there!

Rachel

On Tue, Mar 18, 2025 at 5:45 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi all – An update from my end. I spoke with NOAA about the dive data from last summer's tagging effort. Unfortunately, the data haven't been processed and probably won't be available for quite some time, although it is a science priority for the agency. I asked whether the data are generally consistent with the depth usage reported in Kok et al. 2023 and was told that they are; so I think we should proceed with the approach we've discussed below.

Rachel, don't know if you heard from Tyler about the monthly data?

Also, fyi, Regan and I spoke with Jess and Becca on Friday. They're interested in collaborating on company outreach, which is great news. Either Regan or I will be down next month for the Blue Whales & Blue Skies awards event.

Best,
Michael

From: Jasny, Michael
Sent: Thursday, March 13, 2025 8:01 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

I think that's right—we want to use the same date range that you used in your analysis. Thanks, Rachel.

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, March 13, 2025 7:58 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

I will reach out to Tyler today to see if we can get the remaining AIS data. Double checking the date range for the analysis is the 2023 calendar year?

On Thu, Mar 13, 2025 at 7:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at [la WhaleSafe.com](https://www.whalesafe.com), given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>

Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:



Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

```
vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv
```

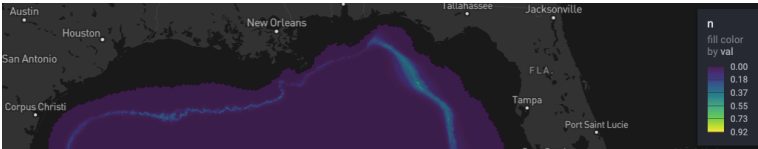
date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461

2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly **n** and **cv** values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

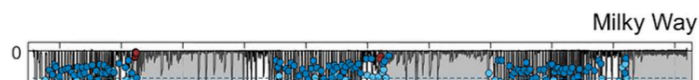


Figure 1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). 21Jan25 with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNU7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

• [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- [github.com/SEFSC/VesselStrikeRiskModel](#)

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model (Kelley et al, 2020) and a couple of R packages to facilitate calculation:

• [shipstrike](#)

Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*

Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*

• [whalestrike](#)

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,

Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

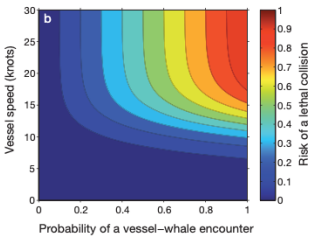
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection. PLOS ONE](#)

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben’s schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!
Michael

MICHAEL JASNY
Director, Marine Mammal Protection

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Please save paper.
Think before printing.

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Fri, Apr 11, 2025 at 3:08 PM

Had a little time to dig into this. Do you still have access to GFW pipelines? If so, Tyler has added the data to this table, 'scratch_benioff_ttl90', within 'world-fishing-827' for us to query.

If you don't have access, I can save daily CSV's from that table for you and upload to Google Drive.

Thanks

[Quoted text hidden]

Mon, Apr 14, 2025 at 1:20 PM

Hi Ben,

I just gave you access to the table in [BigQuery](#). The table includes vessel presence grouped by individual MMSI for 2023. Let me know if you have any issues querying it.

Rachel
[Quoted text hidden]

Mon, Apr 14, 2025 at 2:17 PM

Hi Rachel,

Great, thank you! I don't seem to have access anymore to GFW's world-fishing-827 BigQuery resource (neither as ben@ecoquants.com, nor benbest@ucsb.edu), but can view the linked [BigQuery](#) table you shared.

☐	Field name	Type	Mode
☐	date	DATE	NULLABLE
☐	cell_ll_lat	FLOAT	NULLABLE
☐	cell_ll_lon	FLOAT	NULLABLE
☐	mmsi	STRING	NULLABLE
☐	hours	FLOAT	NULLABLE
☐	dist_travelled_km	FLOAT	NULLABLE
☐	avg_speed_knots	FLOAT	NULLABLE

Do we have another table about **ownership** linking MMSI with the focal companies?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

And for these MMSIs do we have any **characteristics** like vessel length or draft?

Thanks a bunch, Ben

[Quoted text hidden]

Rachel Rhodes <rachelrhodes@ucsb.edu>
To: Ben Best <ben@ecoquants.com>
Cc: "Jasny, Michael" <mjasny@nrdc.org>, "Chen, Leyi" <lchen@nrdc.org>

Mon, Apr 14, 2025 at 3:48 PM

Ok, I just added the **ihs_data_all** table with vessel characteristics to that dataset so you should have access to both tables now. You will probably recall this from Whale Safe days, but you will want to match on **mmsi** and use the most recent **updated_date**. Happy to hop on a quick call this week if you need.

Confirming that as a viewer you can query those tables?

[Quoted text hidden]

Ben Best <ben@ecoquants.com>
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: "Jasny, Michael" <mjasny@nrdc.org>, "Chen, Leyi" <lchen@nrdc.org>

Mon, Apr 14, 2025 at 4:39 PM

Awesome! I see this table **ihs_data_all** now:

▼ benioff-ocean-initiative

▶ 📁 Repositories

▶ 🔍 Saved queries (3)

▶ 🏗️ Pipelines

▶ ➡️ External connections

▼ 📦 gulf_of_mexico_2023

📦 ihs_data_all

📦 vessel_presence_bymmsi_10_v1

I am able to download the tables in chunks (using `bigquery::bq_table_download()`), but I do not have permissions to run queries there (using `bigquery::bq_project_query()`):

```
! Access Denied: Project benioff-ocean-initiative: User does not have
bigquery.jobs.create permission in project benioff-ocean-initiative.
[accessDenied]
```

I'm downloading in 1 GB chunks and saving as parquet files (using `arrow::write_parquet()`); then `arrow::open_dataset()` to read multi-file dataset and eventually with `duckdb`), which I think should work, but open to alternative suggestions.

Cheers, Ben

On Mon, Apr 14, 2025 at 5:49 PM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Ok, I just added the **ihms_data_all** table with vessel characteristics to that dataset so you should have access to both tables now. You will probably recall this from Whale Safe days, but you will want to match on **mmsi** and use the most recent **updated_date**. Happy to hop on a quick call this week if you need.

Confirming that as a viewer you can query those tables?

On Mon, Apr 14, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:
Hi Rachel,

Great, thank you! I don't seem to have access anymore to GFW's world-fishing-827 BigQuery resource (neither as ben@ecoquants.com, nor benbest@ucsb.edu), but can view the linked [BigQuery](#) table you shared.

▼ benioff-ocean-initiative

► Repositories

► Saved queries (3)

► Pipelines

► External connections

▼ gulf_of_mexico_2023

▼ vessel_presence_bymmsi_10_v1

☐	Field name	Type	Mode
☐	date	DATE	NULLABLE
☐	cell_ll_lat	FLOAT	NULLABLE
☐	cell_ll_lon	FLOAT	NULLABLE
☐	mmsi	STRING	NULLABLE
☐	hours	FLOAT	NULLABLE
☐	dist_travelled_km	FLOAT	NULLABLE
☐	avg_speed_knots	FLOAT	NULLABLE

Do we have another table about **ownership** linking MMSI with the focal companies?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

And for these MMSIs do we have any **characteristics** like vessel length or draft?

Thanks a bunch, Ben

On Mon, Apr 14, 2025 at 3:21 PM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:
Hi Ben,

I just gave you access to the table in [BigQuery](#). The table includes vessel presence grouped by individual MMSI for 2023. Let me know if you have any issues querying it.

Rachel

On Thu, Apr 10, 2025 at 8:14 AM Jasny, Michael <mjasny@nrdc.org> wrote:
Thanks so much, Rachel, and to Tyler as well!
Michael

Get [Outlook for iOS](#)

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, April 10, 2025 6:01:56 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice’s whale ship-strike risk

Hi Rachel,

Ok, sounds good! Thanks for the update. Goodluck with the manuscript revisions.

Appreciate all the help, Ben

On Wed, Apr 9, 2025 at 8:10 PM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:
Hi Ben and Michael,

I just got updated tables from Tyler so I can finally pull the raster layers for you. I unfortunately have a competing deadline for manuscript revisions due Friday so I will try my best to get these to you by EOD Friday, but if not you can expect the layers Monday.

Sorry for the delay on this!
Rachel

On Tue, Apr 1, 2025 at 12:08 PM Ben Best <ben@ecoquants.com> wrote:

Hi Rachel and Michael,

I'm ready to start work on this as soon as the ship data can be shared, including AIS time/location and ship company/length/draft where possible.

Rachel, please let me know if there's anything I can do to help with preparing any of this data.

Thanks a bunch, Ben

On Tue, Mar 25, 2025 at 1:06 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Rachel! From our end, we'd like to move forward as quickly as possible, so the timing really depends on Ben's availability.

Ben, do you have a sense of when you can start? Also, please let me know if you have any edits to the contract and scope of work.

Thanks,

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, March 20, 2025 2:29 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi all,

Thanks for these updates. I reached out to Tyler about AIS data. They are planning to make a similar dataset for the full time-series of our data that they can share publicly, which means they need more time to iron out the details of vessel classes and other features they are going to include. In the meantime, they are going to send the query they used so I can manually pull and generate the 2023 rasters that we need for this analysis.

They were going to discuss this week and get back to me with more details. Ideally, when would you like to have AIS data in hand?

That is great news that Jess & Becca were onboard for helping with outreach. I will be at the awards ceremony in Long Beach - it will be great to see you or Regan there!

Rachel

On Tue, Mar 18, 2025 at 5:45 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi all – An update from my end. I spoke with NOAA about the dive data from last summer's tagging effort. Unfortunately, the data haven't been processed and probably won't be available for quite some time, although it is a science priority for the agency. I asked whether the data are generally consistent with the depth usage reported in Kok et al. 2023 and was told that they are; so I think we should proceed with the approach we've discussed below.

Rachel, don't know if you heard from Tyler about the monthly data?

Also, fyi, Regan and I spoke with Jess and Becca on Friday. They're interested in collaborating on company outreach, which is great news. Either Regan or I will be down next month for the Blue Whales & Blue Skies awards event.

Best,

Michael

From: Jasny, Michael
Sent: Thursday, March 13, 2025 8:01 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

I think that's right—we want to use the same date range that you used in your analysis. Thanks, Rachel.

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, March 13, 2025 7:58 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

I will reach out to Tyler today to see if we can get the remaining AIS data. Double checking the date range for the analysis is the 2023 calendar year?

On Thu, Mar 13, 2025 at 7:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler’s outputs for 2023?

In the meantime, I’ll put a contract together for Ben’s work. I’ll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice’s whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice’s whale species and Gulf of Mexico vessels:



Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line

- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

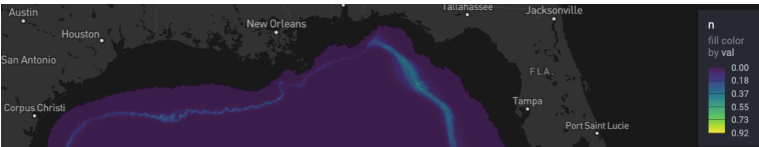
```
vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv
```

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly **n** and **cv** values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

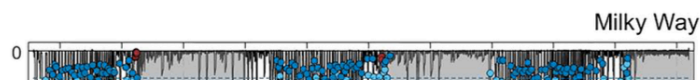


Figure 1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work. 21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

· [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- [shipstrike](#)
 Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
 Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a't First Nation territory](#). *ESR*
- [whalestrike](#)
 Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,

Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

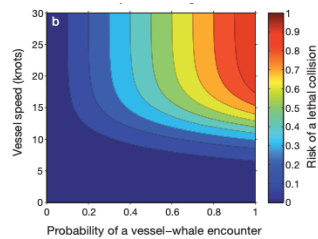
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection.](#) *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben’s schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL
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Please save paper.
Think before printing.

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Mon, Apr 14, 2025 at 4:40 PM

Ben Best <ben@ecoquants.com>
To: "Jasny, Michael" <mjasny@nrdc.org>

Hi Michael.

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at a WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

```
vessel-presence-v1-bymmsi-10-2023-01-01.csv
....
vessel-presence-v1-bymmsi-10-2023-01-31.csv
```

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly μ and cv values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). 21Jan25 with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

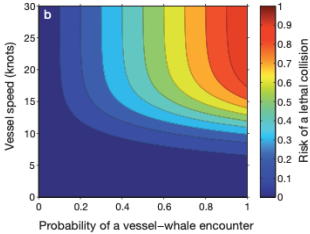
Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
 - Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*
- 
- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM

To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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Jasny, Michael <mjasny@nrdc.org>
To: Ben Best <ben@ecoquants.com>, "Cine, Farrah" <fcine@nrdc.org>

Mon, Apr 14, 2025 at 9:47 PM

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- 1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work, 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

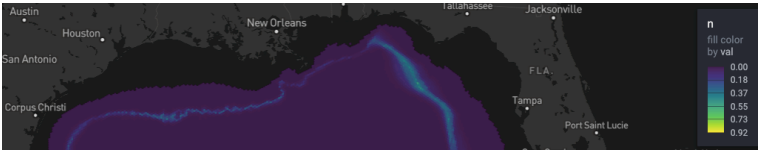
date cell_ll_lat cell_ll_lon mmsi hours dist_travelled_km avg_speed_knots

2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly **n** and **cv** values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

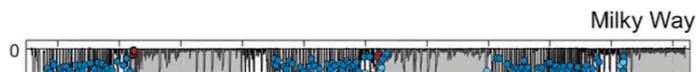


Figure 1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluke cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluke during descent and ascent. Note how Milky Way ceased fluke at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). [21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest

Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

· [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- [shipstrike](#)
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- [whalestrike](#)
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

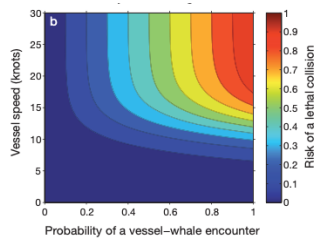
Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
WhaleSafe End Goal: Effect Behavioral Change



1. **Social Incentives**
show best performer.

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#).
Endangered Species Research



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection. PLOS ONE](#)

Blue whale mortality in California



Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!
Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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Please save paper.
Think before printing.

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2 attachments



Rachel Rhodes <rachelrhodes@ucsb.edu>
To: Ben Best <ben@ecoquants.com>
Cc: "Jasny, Michael" <mjasny@nrdc.org>, "Chen, Leyi" <lchen@nrdc.org>

Tue, Apr 15, 2025 at 11:08 AM

Hi Ben,

Sounds like a good download plan to me. I just checked with Chris and he suggested you copy the dataset to your own google cloud project and then you should be able to run any queries on it from there if you need to do anything beyond downloading. Let me know if you have issues or need help.

On Mon, Apr 14, 2025 at 3:40 PM Ben Best <ben@ecoquants.com> wrote:
Awesome! I see this table [ihs_data_all](#) now:

I am able to download the tables in chunks (using `bigquery::bq_table_download()`), but I do not have permissions to run queries there (using `bigquery::bq_project_query()`):

! Access Denied: Project benioff-ocean-initiative: User does not have bigquery.jobs.create permission in project benioff-ocean-initiative.
[accessDenied]

I'm downloading in 1 GB chunks and saving as parquet files (using `arrow::write_parquet()`); then `arrow::open_dataset()` to read multi-file dataset and eventually with `duckdb`), which I think should work, but open to alternative suggestions.

Cheers, Ben

On Mon, Apr 14, 2025 at 5:49 PM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Ok, I just added the `lhs_data_all` table with vessel characteristics to that dataset so you should have access to both tables now. You will probably recall this from Whale Safe days, but you will want to match on `mmsi` and use the most recent `updated_date`. Happy to hop on a quick call this week if you need.

Confirming that as a viewer you can query those tables?

On Mon, Apr 14, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Hi Rachel,

Great, thank you! I don't seem to have access anymore to GFW's world-fishing-827 BigQuery resource (neither as ben@ecoquants.com, nor benbest@ucsb.edu), but can view the linked [BigQuery](#) table you shared.

Do we have another table about **ownership** linking MMSI with the focal companies?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd

- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

And for these MMSIs do we have any **characteristics** like vessel length or draft?

Thanks a bunch, Ben

On Mon, Apr 14, 2025 at 3:21 PM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben,

I just gave you access to the table in [BigQuery](#). The table includes vessel presence grouped by individual MMSI for 2023. Let me know if you have any issues querying it.

Rachel

On Thu, Apr 10, 2025 at 8:14 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks so much, Rachel, and to Tyler as well!
Michael

Get [Outlook for iOS](#)

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, April 10, 2025 6:01:56 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Rachel,

Ok, sounds good! Thanks for the update. Goodluck with the manuscript revisions.

Appreciate all the help, Ben

On Wed, Apr 9, 2025 at 8:10 PM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I just got updated tables from Tyler so I can finally pull the raster layers for you. I unfortunately have a competing deadline for manuscript revisions due Friday so I will try my best to get these to you by EOD Friday, but if not you can expect the layers Monday.

Sorry for the delay on this!
Rachel

On Tue, Apr 1, 2025 at 12:08 PM Ben Best <ben@ecoquants.com> wrote:

Hi Rachel and Michael,

I'm ready to start work on this as soon as the ship data can be shared, including AIS time/location and ship company/length/draft where possible.

Rachel, please let me know if there's anything I can do to help with preparing any of this data.

Thanks a bunch, Ben

On Tue, Mar 25, 2025 at 1:06 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Rachel! From our end, we'd like to move forward as quickly as possible, so the timing really depends on Ben's availability.

Ben, do you have a sense of when you can start? Also, please let me know if you have any edits to the contract and scope of work.

Thanks,

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, March 20, 2025 2:29 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi all,

Thanks for these updates. I reached out to Tyler about AIS data. They are planning to make a similar dataset for the full time-series of our data that they can share publicly, which means they need more time to iron out the details of vessel classes and other features they are going to include. In the meantime, they are going to send the query they used so I can manually pull and generate the 2023 rasters that we need for this analysis.

They were going to discuss this week and get back to me with more details. Ideally, when would you like to have AIS data in hand?

That is great news that Jess & Becca were onboard for helping with outreach. I will be at the awards ceremony in Long Beach - it will be great to see you or Regan there!

Rachel

On Tue, Mar 18, 2025 at 5:45 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi all – An update from my end. I spoke with NOAA about the dive data from last summer's tagging effort. Unfortunately, the data haven't been processed and probably won't be available for quite some time, although it is a science priority for the agency. I asked whether the data are generally consistent with the depth usage reported in Kok et al. 2023 and was told that they are; so I think we should proceed with the approach we've discussed below.

Rachel, don't know if you heard from Tyler about the monthly data?

Also, fyi, Regan and I spoke with Jess and Becca on Friday. They're interested in collaborating on company outreach, which is great news. Either Regan or I will be down next month for the Blue Whales & Blue Skies awards event.

Best,

Michael

From: Jasny, Michael
Sent: Thursday, March 13, 2025 8:01 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

I think that's right—we want to use the same date range that you used in your analysis. Thanks, Rachel.

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, March 13, 2025 7:58 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

I will reach out to Tyler today to see if we can get the remaining AIS data. Double checking the date range for the analysis is the 2023 calendar year?

On Thu, Mar 13, 2025 at 7:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at a WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:



Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

```
vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv
```

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit [shiny.marinesensitivity.org/sdm](#) to see monthly n and cv values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

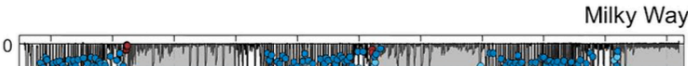


Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). [21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

· [whalestrike](#)Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- [github.com/SEFSC/VesselStrikeRiskModel](#)

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- [shipstrike](#)
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a First Nation territory](#). *ESR*
- [whalestrike](#)
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

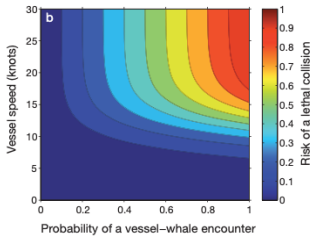
Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
[WhaleSafe End Goal: Effect Behavioral Change](#)



1. Social Incentives

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben’s schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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Please save paper.
Think before printing.

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Cine, Farrah <fcine@nrdc.org>
To: "Jasny, Michael" <mjasny@nrdc.org>, Ben Best <ben@ecoquants.com>

Wed, Apr 16, 2025 at 8:38 AM

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,
Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think Kok et al (2023) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the Blondin et al (2025) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tyler's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly μ and cv values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1 . Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work. 21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast. *Scientific Reports*](#)
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates. *Ocean & Coastal Management*](#)
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory. *ESR*](#)
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk. *Biological Conservation*](#)

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

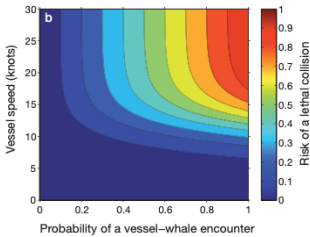
On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

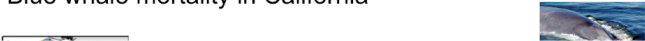
Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Blue whale mortality in California



Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM

To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

4479 W. 5TH AVENUE

VANCOUVER, BC V6R1S4

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MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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Ben Best <ben@ecoquants.com>
To: "Cine, Farrah" <fcine@nrdc.org>
Cc: "Jasny, Michael" <mjasny@nrdc.org>

Tue, Apr 22, 2025 at 11:41 AM

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

[Quoted text hidden]

Cine, Farrah <fcine@nrdc.org>
To: Ben Best <ben@ecoquants.com>
Cc: "Jasny, Michael" <mjasny@nrdc.org>

Tue, Apr 22, 2025 at 11:45 AM

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>

[Quoted text hidden]

[Quoted text hidden]

Jasny, Michael <mjasny@nrdc.org>
To: "Cine, Farrah" <fcine@nrdc.org>, Ben Best <ben@ecoquants.com>

Tue, Apr 22, 2025 at 11:54 AM

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."



Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per Blondin et al (2025) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 (Garrison et al, 2025) with code found at SEFSC/VesselStrikeRiskModel. I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:



Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUICruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv

...

vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly μ and σ values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:



Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). 21Jan25 with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- **whalestrike**
 Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

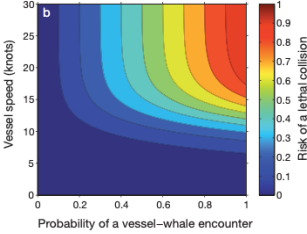
Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
WhaleSafe End Goal: Effect Behavioral Change
 - Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*
- 
- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM

To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

4479 W. 5TH AVENUE

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C 310.560.5536
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Please save paper.
Think before printing.

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Environmental Data Scientist

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Ben Best <ben@ecoquants.com>
To: "Jasny, Michael" <mjasny@nrdc.org>
Cc: "Cine, Farrah" <fcine@nrdc.org>

Tue, Apr 22, 2025 at 4:32 PM

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice’s whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Day	% Time	Night	Day	% Time	Night
20	0	20	20	0	20

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:



Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work, 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date cell_ll_lat cell_ll_lon mmsi hours dist_travelled_km avg_speed_knots

2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly μ and cv values



As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Milky Way

Figure 1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). 21 Jan 25 with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEFNU7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <chen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

· [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

• [shipstrike](#)

Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*

Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*

• [whalestrike](#)

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,

Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

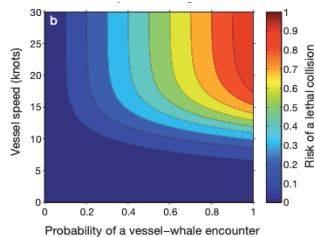
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
 WhaleSafe End Goal: Effect Behavioral Change

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf. Endangered Species Research](#)



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection. PLOS ONE](#)

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben’s schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!
Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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Please save paper.
Think before printing.

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Jasny, Michael <mjasny@nrdc.org>
To: Ben Best <ben@ecoquants.com>, "Cine, Farrah" <fcine@nrdc.org>

Mon, Apr 28, 2025 at 2:09 PM

Hi Farrah – Just checking on progress on Ben's contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the [Blondin et al. \(2025\)](#)/ [Garrison et al. \(2025\)](#) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- 1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102

2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly **n** and **cv** values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased

fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work. 21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a't First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United](#)

States East Coast. Scientific Reports

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk. Biological Conservation](#)

The actual references for this R package include the biophysical model previously mentioned:

· [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

• [shipstrike](#)

Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*

Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*

• [whalestrike](#)

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,

Rachel

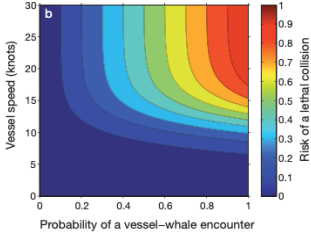
On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*
[Blue whale mortality in California](#)

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben’s schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

4479 W. 5TH AVENUE

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Please save paper.
Think before printing.

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Ben Best <ben@ecoquants.com>
To: "Jasny, Michael" <mjasny@nrdc.org>
Cc: "Cine, Farrah" <fcine@nrdc.org>, Lauren Wilkins <lwilkins@nrdc.org>

Tue, Apr 29, 2025 at 5:55 AM

Hi Michael,

I just signed the DocuSign that Lauren kindly updated with my latest business address.

I'll work on this project again starting Friday.

Many thanks, Ben

On Mon, Apr 28, 2025 at 10:09 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Farrah – Just checking on progress on Ben's contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny ($n=2$), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the [Blondin et al. \(2025\)](#)/[Garrison et al. \(2025\)](#) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. OK, I think this is the right approach to target operators directly at a WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Miridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv

...

vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit [shiny.marinesensitivity.org/sdm](#) to see monthly μ and cv values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFN7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- [whalestrike](#)
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a First Nation territory](#). *ESR*
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

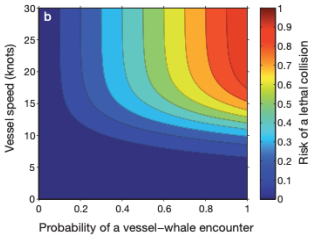
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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Jasny, Michael <mjasny@nrdc.org>
To: Ben Best <ben@ecoquants.com>
Cc: "Cine, Farrah" <fcine@nrdc.org>, "Wilkins, Lauren" <lwilkins@nrdc.org>

Tue, Apr 29, 2025 at 6:43 AM

Thanks, Ben! Please let me know when you have an ETA. Is May possible?

Lots of positive developments. The Blue Whales Blue Skies folks will be partnering on shipping company outreach. And relatedly, a couple of major ports (Corpus Cristi and Galveston) have begun expressing interest in noise reduction measures.

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 29, 2025 4:56 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

I just signed the DocuSign that Lauren kindly updated with my latest business address.

I'll work on this project again starting Friday.

Many thanks, Ben

On Mon, Apr 28, 2025 at 10:09 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Farrah – Just checking on progress on Ben's contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

```
vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv
```

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291
2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly **n** and **cv** values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). [21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
 Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
 Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
 Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
 Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
 Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

• [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

• [shipstrike](#)

Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*

Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*

• [whalestrike](#)

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,

Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

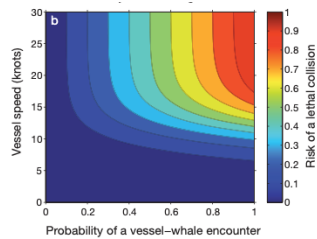
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection. PLOS ONE](#)

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben’s schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!
Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

4479 W. 5TH AVENUE

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C 310.560.5536
M.JASNY@NRDC.ORG

Please save paper.
Think before printing.

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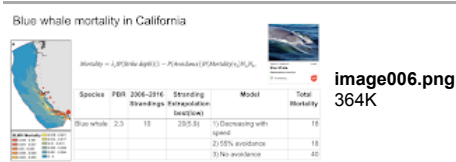
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Ben Best <ben@ecoquants.com>

To: "Jasny, Michael" <mjasny@nrdc.org>

Cc: "Cine, Farrah" <fcine@nrdc.org>, "Wilkins, Lauren" <lwilkins@nrdc.org>

Wed, Apr 30, 2025 at 3:48 AM

Hi Michael,

That's all good news! Yes, by the end of May, if not sooner.

Cheers, Ben

On Tue, Apr 29, 2025 at 2:43 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! Please let me know when you have an ETA. Is May possible?

Lots of positive developments. The Blue Whales Blue Skies folks will be partnering on shipping company outreach. And relatedly, a couple of major ports (Corpus Cristi and Galveston) have begun expressing interest in noise reduction measures.

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 29, 2025 4:56 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice’s whale ship-strike risk

Hi Michael,

I just signed the DocuSign that Lauren kindly updated with my latest business address.

I'll work on this project again starting Friday.

Many thanks, Ben

On Mon, Apr 28, 2025 at 10:09 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Farrah – Just checking on progress on Ben’s contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice’s whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM

To: Jasny, Michael <mjasny@nrdc.org>

Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>

Sent: Friday, March 28, 2025 9:21 AM

To: Jasny, Michael <mjasny@nrdc.org>

Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>

Sent: Wednesday, March 26, 2025 3:59 PM

To: Jasny, Michael <mjasny@nrdc.org>

Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the [Blondin et al. \(2025\)](#)/ [Garrison et al. \(2025\)](#) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- 1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit [shiny.marinesensitivity.org/sdm](#) to see monthly μ and σ values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). [21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- [whalestrike](#)
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- [github.com/SEFSC/VesselStrikeRiskModel](#)

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model (Kelley et al, 2020) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>

Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

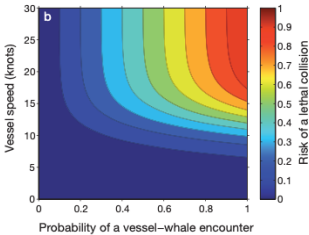
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <chen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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Please save paper.
Think before printing.

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Jasny, Michael <mjasny@nrdc.org>
To: Ben Best <ben@ecoquants.com>

Mon, Jun 23, 2025 at 12:54 PM

Hi Ben – Hope the summer is going well! I'm just writing to check on the risk-reduction analysis for Rice's whale. How has it gone, and do you have an ETA? Relatedly, I'll need an invoice by the end of the month in order to budget the work against our current fiscal year.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, April 30, 2025 2:48 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

That's all good news! Yes, by the end of May, if not sooner.

Cheers, Ben

On Tue, Apr 29, 2025 at 2:43 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! Please let me know when you have an ETA. Is May possible?

Lots of positive developments. The Blue Whales Blue Skies folks will be partnering on shipping company outreach. And relatedly, a couple of major ports (Corpus Cristi and Galveston) have begun expressing interest in noise reduction measures.

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 29, 2025 4:56 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

I just signed the DocuSign that Lauren kindly updated with my latest business address.

I'll work on this project again starting Friday.

Many thanks, Ben

On Mon, Apr 28, 2025 at 10:09 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Farrah – Just checking on progress on Ben's contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>

Cc: Jasny, Michael <mjasny@nrdc.org>

Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>

Sent: Tuesday, April 22, 2025 1:41 PM

To: Cine, Farrah <fcine@nrdc.org>

Cc: Jasny, Michael <mjasny@nrdc.org>

Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>

Sent: Monday, April 14, 2025 11:47 PM

To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>

Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,

Michael

From: Ben Best <ben@ecoquants.com>

Sent: Monday, April 14, 2025 3:41 PM

To: Jasny, Michael <mjasny@nrdc.org>

Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the [Blondin et al. \(2025\)](#)/ [Garrison et al. \(2025\)](#) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- 1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25?](#)

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUICruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit shiny.marinesensitivity.org/sdm to see monthly μ and cv values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). [21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
 Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
 Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
 Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
 Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
 Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEFNU7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- [whalestrike](#)
- Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- [github.com/SEFSC/VesselStrikeRiskModel](#)

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model (Kelley et al, 2020) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>

Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

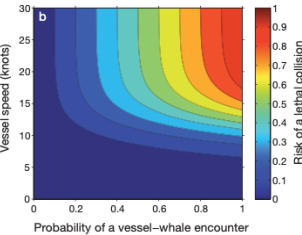
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

[4479 W. 5TH AVENUE](#)

[VANCOUVER, BC V6R1S4](#)

C 310.560.5536
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Please save paper.
Think before printing.

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2 attachments



image005.png
514K



image006.png
364K

Ben Best <ben@ecoquants.com>
To: "Jasny, Michael" <mjasny@nrdc.org>

Hi Michael,

Thank you for reaching out. My apologies for not doing so sooner. I got consumed working on a project defunded by Trump but is now back online, and hopefully for another 4 years.

I've downloaded the > 53 GB of ship presence data provided by Tylar at GFW and begun processing it. There are some spikes in time and a sliver of a gap in space I will ask him about. Meanwhile, can you please share the shapefile for the 7 subareas?

I'm going to try to get final results to you by the end of the month.

Many thanks, Ben

PS Here's a screenshot of the overall coverage in space:

and time:

On Mon, Jun 23, 2025 at 8:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Hope the summer is going well! I'm just writing to check on the risk-reduction analysis for Rice's whale. How has it gone, and do you have an ETA? Relatedly, I'll need an invoice by the end of the month in order to budget the work against our current fiscal year.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, April 30, 2025 2:48 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

That's all good news! Yes, by the end of May, if not sooner.

Cheers, Ben

On Tue, Apr 29, 2025 at 2:43 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! Please let me know when you have an ETA. Is May possible?

Lots of positive developments. The Blue Whales Blue Skies folks will be partnering on shipping company outreach. And relatedly, a couple of major ports (Corpus Cristi and Galveston) have begun expressing interest in noise reduction measures.

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 29, 2025 4:56 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

I just signed the DocuSign that Lauren kindly updated with my latest business address.

I'll work on this project again starting Friday.

Many thanks, Ben

On Mon, Apr 28, 2025 at 10:09 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Farrah – Just checking on progress on Ben's contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here’s the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I’ve revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I’ve also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I’ll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice’s whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice’s Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice’s Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per Blondin et al (2025) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 (Garrison et al, 2025) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUICruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

```
vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv
```

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit [shiny.marinesensitivity.org/sdm](#) to see monthly μ and σ values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure 1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). [21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <chen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

1. **simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
2. **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
3. **whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
4. **VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEFNU7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- [whalestrike](#)
- Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- [shipstrike](#)
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a't First Nation territory](#). *ESR*
- [whalestrike](#)
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>

Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

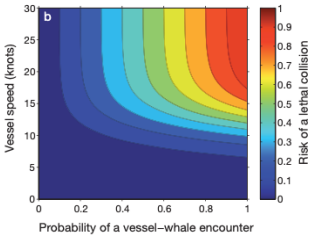
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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C 310.560.5536

MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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Jasny, Michael <mjasny@nrdc.org>
To: Ben Best <ben@ecoquants.com>, "Chen, Leyi" <lchen@nrdc.org>

Tue, Jun 24, 2025 at 10:54 AM

Hi Leyi – Could you forward ASAP the shapefiles for our 7 Gulf of Mexico subareas (which you developed for the Benioff vessel analysis) to Ben, copied?

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, June 24, 2025 9:37 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Thank you for reaching out. My apologies for not doing so sooner. I got consumed working on a project defunded by Trump but is now back online, and hopefully for another 4 years.

I've downloaded the > 53 GB of ship presence data provided by Tylar at GFW and begun processing it. There are some spikes in time and a sliver of a gap in space I will ask him about. Meanwhile, can you please share the shapefile for the 7 subareas?

I'm going to try to get final results to you by the end of the month.

Many thanks, Ben

PS Here's a screenshot of the overall coverage in space:

and time:

On Mon, Jun 23, 2025 at 8:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Hope the summer is going well! I'm just writing to check on the risk-reduction analysis for Rice's whale. How has it gone, and do you have an ETA? Relatedly, I'll need an invoice by the end of the month in order to budget the work against our current fiscal year.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, April 30, 2025 2:48 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

That's all good news! Yes, by the end of May, if not sooner.

Cheers, Ben

On Tue, Apr 29, 2025 at 2:43 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! Please let me know when you have an ETA. Is May possible?

Lots of positive developments. The Blue Whales Blue Skies folks will be partnering on shipping company outreach. And relatedly, a couple of major ports (Corpus Cristi and Galveston) have begun expressing interest in noise reduction measures.

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 29, 2025 4:56 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

I just signed the DocuSign that Lauren kindly updated with my latest business address.

I'll work on this project again starting Friday.

Many thanks, Ben

On Mon, Apr 28, 2025 at 10:09 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Farrah – Just checking on progress on Ben's contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv

...

vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit [shiny.marinesensitivity.org/sdm](#) to see monthly μ and cv values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work. 21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

- simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
- shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
- VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- [whalestrike](#)
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model (Kelley et al, 2020) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a First Nation territory](#). *ESR*
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>

Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16AM Ben Best <ben@ecoquants.com> wrote:

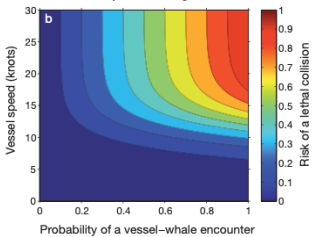
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

[4479 W. 5TH AVENUE](#)

[VANCOUVER, BC V6R1S4](#)

C 310.560.5536
MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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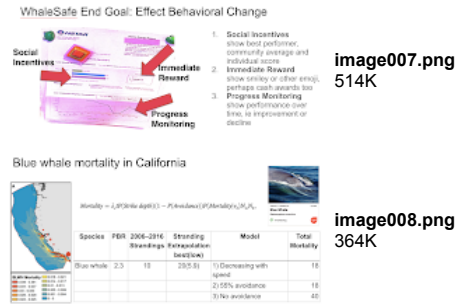
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Environmental Data Scientist

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2 attachments



Chen, Leyi <lchen@nrdc.org>

To: "Jasny, Michael" <mjasny@nrdc.org>, Ben Best <ben@ecoquants.com>

Tue, Jun 24, 2025 at 11:22 AM

Hi Michael,

No problem! I'm attaching the shapefile here.

Best,
Leyi

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Tuesday, June 24, 2025 12:54 PM
To: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: RE: Further analysis of Rice’s whale ship-strike risk

Hi Leyi – Could you forward ASAP the shapefiles for our 7 Gulf of Mexico subareas (which you developed for the Benioff vessel analysis) to Ben, copied?

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, June 24, 2025 9:37 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice’s whale ship-strike risk

Hi Michael,

Thank you for reaching out. My apologies for not doing so sooner. I got consumed working on a project defunded by Trump but is now back online, and hopefully for another 4 years.

I've downloaded the > 53 GB of ship presence data provided by Tylar at GFW and begun processing it. There are some spikes in time and a sliver of a gap in space I will ask him about. Meanwhile, can you please share the shapefile for the 7 subareas?

I'm going to try to get final results to you by the end of the month.

Many thanks, Ben

PS Here's a screenshot of the overall coverage in space:

and time:

On Mon, Jun 23, 2025 at 8:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Hope the summer is going well! I'm just writing to check on the risk-reduction analysis for Rice's whale. How has it gone, and do you have an ETA? Relatedly, I'll need an invoice by the end of the month in order to budget the work against our current fiscal year.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, April 30, 2025 2:48 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

That's all good news! Yes, by the end of May, if not sooner.

Cheers, Ben

On Tue, Apr 29, 2025 at 2:43 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! Please let me know when you have an ETA. Is May possible?

Lots of positive developments. The Blue Whales Blue Skies folks will be partnering on shipping company outreach. And relatedly, a couple of major ports (Corpus Cristi and Galveston) have begun expressing interest in noise reduction measures.

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 29, 2025 4:56 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

I just signed the DocuSign that Lauren kindly updated with my latest business address.

I'll work on this project again starting Friday.

Many thanks, Ben

On Mon, Apr 28, 2025 at 10:09 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Farrah – Just checking on progress on Ben's contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per Blondin et al (2025) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 (Garrison et al, 2025) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- 1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit [shiny.marinesensitivity.org/sdm](#) to see monthly μ and cv values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work. 21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

- simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
- shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
- VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10**: Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10**: Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

- [whalestrike](#)
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model (Kelley et al, 2020) and a couple of R packages to facilitate calculation:

- **shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'a First Nation territory](#). *ESR*
- **whalestrike**
Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>

Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,
Rachel

On Wed, Nov 6, 2024 at 9:16AM Ben Best <ben@ecoquants.com> wrote:

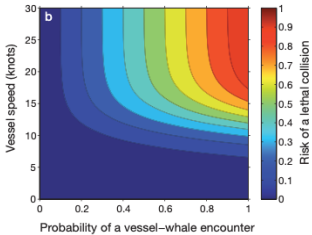
Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

- Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*



- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we're on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn't one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben's schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY
Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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[VANCOUVER, BC V6R1S4](#)

[C 310.560.5536](#)
MJASNY@NRDC.ORG

Please save paper.
Think before printing.

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 **Rice's Whale Proposed Critical Habitat Sub-Areas.zip**
49K

Ben Best <ben@ecoquants.com>
To: "Chen, Leyi" <lchen@nrdc.org>
Cc: "Jasny, Michael" <mjasny@nrdc.org>

This is perfect, thank you!

On Tue, Jun 24, 2025 at 7:22 PM Chen, Leyi <lchen@nrdc.org> wrote:

Hi Michael,

No problem! I'm attaching the shapefile here.

Best,
Leyi

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Tuesday, June 24, 2025 12:54 PM
To: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Leyi – Could you forward ASAP the shapefiles for our 7 Gulf of Mexico subareas (which you developed for the Benioff vessel analysis) to Ben, copied?

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, June 24, 2025 9:37 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Thank you for reaching out. My apologies for not doing so sooner. I got consumed working on a project defunded by Trump but is now back online, and hopefully for another 4 years.

I've downloaded the > 53 GB of ship presence data provided by Tylar at GFW and begun processing it. There are some spikes in time and a sliver of a gap in space I will ask him about. Meanwhile, can you please share the shapefile for the 7 subareas?

I'm going to try to get final results to you by the end of the month.

Many thanks, Ben

PS Here's a screenshot of the overall coverage in space:

and time:

On Mon, Jun 23, 2025 at 8:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Hope the summer is going well! I'm just writing to check on the risk-reduction analysis for Rice's whale. How has it gone, and do you have an ETA? Relatedly, I'll need an invoice by the end of the month in order to budget the work against our current fiscal year.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, April 30, 2025 2:48 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

That's all good news! Yes, by the end of May, if not sooner.

Cheers, Ben

On Tue, Apr 29, 2025 at 2:43 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! Please let me know when you have an ETA. Is May possible?

Lots of positive developments. The Blue Whales Blue Skies folks will be partnering on shipping company outreach. And relatedly, a couple of major ports (Corpus Cristi and Galveston) have begun expressing interest in noise reduction measures.

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 29, 2025 4:56 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

I just signed the DocuSign that Lauren kindly updated with my latest business address.

I'll work on this project again starting Friday.

Many thanks, Ben

On Mon, Apr 28, 2025 at 10:09 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Farrah – Just checking on progress on Ben's contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice’s whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly at WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

- 1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
- 3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUI Cruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv
...
vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit [shiny.marinesensitivity.org/sdm](#) to see monthly **n** and **cv** values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work](#). [21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

- 1. simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
- 2. shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- 3. whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
- 4. VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-oaLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

· [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
 Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- [shipstrike](#)
 Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
 Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- [whalestrike](#)
 Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,

Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:

 - Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*
-
- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben’s schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

MICHAEL JASNY

Director, Marine Mammal Protection

NATURAL RESOURCES DEFENSE COUNCIL

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Please save paper.
Think before printing.

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Ben Best <ben@ecoquants.com>
To: "Jasny, Michael" <mjasny@nrdc.org>

Hi Michael,

I'm not done yet, and I know you're end of the year deadline is today. So far I've spent 21 hrs (\$3,780). Do you need an invoice now, or can you wait until the end of the week?

I read in all the data and converted to the same 10th degree grid (including NMFS critical habitat and subareas) as the vessel presence, but need more time working through the [VesselStrikeRiskModel](#). I managed to digitize the needed residency parameters from the only known source [Kok et al \(2023\)](#) to the 5 m and 15 m draft depths, but I think it's critical to split out results for night and day (which is not in the N Atlantic right whale [VesselStrikeRiskModel](#)), per Figure 5:

[Kok et al \(2023\)](#) **Figure 5.** Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).
Table of digitized cumulative time spent at or above 5 and 15 m depth:

depth_m	pct_time	period	whale
5	33%	day	2015
5	33%	day	2018
5	65%	night	2015
5	72%	night	2018
15	41%	day	2015
15	39%	day	2018
15	85%	night	2015
15	85%	night	2018

Thanks a bunch, Ben

On Wed, Jun 25, 2025 at 5:24 PM Ben Best <ben@ecoquants.com> wrote:

This is perfect, thank you!

On Tue, Jun 24, 2025 at 7:22 PM Chen, Leyi <lchen@nrdc.org> wrote:

Hi Michael,

No problem! I'm attaching the shapefile here.

Best,
Leyi

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Tuesday, June 24, 2025 12:54 PM
To: Ben Best <ben@ecoquants.com>; Chen, Leyi <lchen@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Leyi – Could you forward ASAP the shapefiles for our 7 Gulf of Mexico subareas (which you developed for the Benioff vessel analysis) to Ben, copied?

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, June 24, 2025 9:37 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Thank you for reaching out. My apologies for not doing so sooner. I got consumed working on a project defunded by Trump but is now back online, and hopefully for another 4 years.

I've downloaded the > 53 GB of ship presence data provided by Tylar at GFW and begun processing it. There are some spikes in time and a sliver of a gap in space I will ask him about. Meanwhile, can you please share the shapefile for the 7 subareas?

I'm going to try to get final results to you by the end of the month.

Many thanks, Ben

PS Here's a screenshot of the overall coverage in space:

and time:

On Mon, Jun 23, 2025 at 8:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Hope the summer is going well! I'm just writing to check on the risk-reduction analysis for Rice's whale. How has it gone, and do you have an ETA? Relatedly, I'll need an invoice by the end of the month in order to budget the work against our current fiscal year.

Thanks!
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, April 30, 2025 2:48 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

That's all good news! Yes, by the end of May, if not sooner.

Cheers, Ben

On Tue, Apr 29, 2025 at 2:43 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! Please let me know when you have an ETA. Is May possible?

Lots of positive developments. The Blue Whales Blue Skies folks will be partnering on shipping company outreach. And relatedly, a couple of major ports (Corpus Cristi and Galveston) have begun expressing interest in noise reduction measures.

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 29, 2025 4:56 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>; Wilkins, Lauren <lwilkins@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

I just signed the DocuSign that Lauren kindly updated with my latest business address.

I'll work on this project again starting Friday.

Many thanks, Ben

On Mon, Apr 28, 2025 at 10:09 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Farrah – Just checking on progress on Ben's contract. Have we resolved the DocuSign snafu?

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 3:33 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Cine, Farrah <fcine@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Thank you Farrah and Michael. No worries.

Much appreciated, Ben

On Tue, Apr 22, 2025 at 1:54 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Apologies, Ben. Not sure what the problem could be with the DocuSign.

From: Cine, Farrah <fcine@nrdc.org>
Sent: Tuesday, April 22, 2025 10:45 AM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Ben,

Pinged the reviewer and will respond back once I get an answer for you!

Thanks for your patience.

-Farrah

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, April 22, 2025 1:41 PM
To: Cine, Farrah <fcine@nrdc.org>
Cc: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Farrah,

Is there any progress with sending the DocuSign agreement?

Thanks, Ben

On Wed, Apr 16, 2025 at 10:39 AM Cine, Farrah <fcine@nrdc.org> wrote:

Hi Michael/Ben,

This has been approved. Awaiting the DocuSign being prepared.

Thanks,

Farrah

From: Jasny, Michael <mjasny@nrdc.org>
Sent: Monday, April 14, 2025 11:47 PM
To: Ben Best <ben@ecoquants.com>; Cine, Farrah <fcine@nrdc.org>
Subject: RE: Further analysis of Rice's whale ship-strike risk

Hi Farrah – Can you check on the status of our contract with Ben Best (copied here)? CC may need some pushing. Thanks,
Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, April 14, 2025 3:41 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

FYI, I don't yet see a DocuSign document.

Thanks, Ben

On Mon, Mar 31, 2025 at 2:57 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben. I'll upload it into our system today, and you'll get a message from DocuSign within the next week or two.

Are you still waiting for some data from Rachel/ Tyler?

Also, please let me know your rough sense of timing for deliverables. We can set an informal deadline if helpful.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Friday, March 28, 2025 9:21 AM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Yes, that sounds good! Please find signed contract attached with tracked changes, including new permanent EcoQuants LLC address.

Thanks so much, Ben

On Thu, Mar 27, 2025 at 12:21 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – We tend to use \$100 as a standard rate, but we certainly can go with \$180/hr. Practically speaking, I would like to keep the total to 8K if we can, just given the demands that Trump defense is making on our discretionary budget.

I've reattached the consulting agreement with the numbers changed in the compensation para., so it now reads: "NRDC will pay Consultant at the rate of \$180/hour, up to a total of \$8,000 unless a higher total amount is agreed to by Consultant and NRDC in writing." Let me know if that's ok.

The agreement still has your Santa Barbara address. Is that the one to use, or would you prefer the Atlanta one? We can always switch to your Atlanta address for your invoice(s).

Finally, it would be helpful for our outreach planning to have a target date for deliverables, even if we don't include it in the contract.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 26, 2025 3:59 PM
To: Jasny, Michael <mjasny@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

Apologies for not responding sooner about the contract. We moved out of California for the summer and are in Atlanta this week while my wife shoots a documentary -- my first time caring for Liam alone all day.

My rate has been \$180 per hour for several years (without any inflation adjustment), but the contract says \$100 per hour. Is this negotiable? I originally suggested this would take 60 hrs * \$180 = \$10,800, but I did buffer from what I think it will actually take (40 hrs * \$180 = \$7,200).

The Exhibit A does an excellent job laying out the work to date and what's ahead for this contract. I look forward to digging in.

Much appreciated, Ben

PS Although the sample size is tiny (n=2), I think [Kok et al \(2023\)](#) summarizes the residence time at the surface (ie exposure to ships) vs depth by day vs night that we need to plug into the [Blondin et al \(2025\)](#) model:

"During nighttime, both whales were within 15 m of the sea surface 85% of the time and within 2 m of the surface 45% (Milky Way) and 52% (Edna) of the time (Fig. 5)."

Figure 5. Percent of time spent at depths 0–20 m during daylight (red) and nighttime (blue) for Milky Way (2015) and Edna (2018), along with cumulative time spent at a given depth or shallower (dotted lines).

On Thu, Mar 13, 2025 at 10:38 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben, and agreed!

Here's the contract, with the statement of work included as an appendix. Please look particularly at paragraph (4) in the analysis section of the statement, as I've revised it, in light of our recent emails, to prefer use of the Blondin et al. (2025)/ Garrison et al. (2025) approach if the relevant data are available. I've also added an item on risk-reduction from nighttime transits to para. (5) of the deliverables section (again, if data are available).

Let me know if you have edits. I'll upload the final version into our system, and it will come back to you via Docusign within the next two weeks.

Thanks,

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Wednesday, March 12, 2025 4:11 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: Further analysis of Rice's whale ship-strike risk

Hi Michael,

No problem. Ok, I think this is the right approach to target operators directly a la WhaleSafe.com, given the unfortunate rescinding of protections by BOEM:

- [BOEM Rescinds Expanded Rice's Whale Protection Efforts | Bureau of Ocean Energy Management](#)
- [Defenders Statement on BOEM Rescission of Rice's Whale Recommendations | Defenders of Wildlife](#)

I look forward to collecting the rest of the information needed and finalizing the contract to proceed.

Many thanks, Ben

On Wed, Mar 12, 2025 at 11:47 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben – Apologies for my delay as well!

It looks like we have most if not all the model inputs covered, but please let me know if I'm missing something.

- Vessel characteristics: The key determinant in the Blondin et al. (2025) model is vessel length, and we have that information in the metadata, including for all the companies listed in the statement of work. I've attached the 2023 individual vessel table that Rachel and her team produced, which is searchable by operator. So I think we're good there.
- Vessel operations: It sounds like we need Tyler's monthly outputs for the remainder of 2023. Rachel, can that be provided?
- Whale behavior: I agree that we need to use the tagging depth data from Kok et al. (2023). Soldevilla et al. (2017), attached, published dive data for one whale tagged in 2015, but I'm pretty sure that's just the Milky Way data presented in Kok et al. The only question, then, is whether NMFS would provide data from last summer's research cruise, which used drones to place tags on seven Rice's whales. I'm guessing that those 2024 tagging data won't be available anytime soon and that we'll need to stick with the information in Kok et al., but I can ask Mridula about the new data before a call we're both on this Friday.

Ben, the cost and timing would work for us. I can let you know on Friday if there's a reasonable chance of accessing the new tagging data.

Rachel, could you provide the remainder of Tyler's outputs for 2023?

In the meantime, I'll put a contract together for Ben's work. I'll send it over later today for review.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, February 17, 2025 5:40 PM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Further analysis of Rice's whale ship-strike risk

Hi Michael, Rachel and Leyi,

Apologies for not responding sooner. The best and most modern of the ship strike risk approaches is per [Blondin et al \(2025\)](#) for the N Atlantic right whale, which uses the updated probability of lethal strike risk curve published Feb 5, 2025 ([Garrison et al, 2025](#)) with code found at [SEFSC/VesselStrikeRiskModel](#). I am still unsure how much of this information we can know or estimate of the Rice's whale species and Gulf of Mexico vessels:

Fig. 1. Conceptual diagram outlining the key components of the vessel strike encounter risk model. Model inputs, model parameters, and model outputs are grouped together. Colors are representative of specific model parameters (e.g., model inputs that are in green boxes are used to calculate the encounter rate). Vessel speed and vessel draft are in orange because they are used in the calculation of multiple parameters. Mortality only equals 1 when a strike is equal to 1, the probability of avoidance is equal to 0, and the probability of lethality is equal to 1. Source: [Blondin et al \(2025\)](#).

Based on the code [SEFSC/VesselStrikeRiskModel](#), the vessel "Draft" and "Propellor suction depth" is given by its class ([Garrison et al, 2025](#)):

1. Small/Medium vessels
26 to 65 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
2. Large vessels
65 to 350 feet in length (mean.draft: 5; PropSuctDepth: 1*mean.draft)
3. Ocean-going vessels (OGVs)
over 350 in length (mean.draft: 15; PropSuctDepth: 2*mean.draft)

Do we have length per MMSI and company listed in [Statement of work. 21Jan25](#)?

- CMA CGM
- Stolt Tankers
- MSC
- Maersk
- Hapag-Lloyd
- Carnival Cruise Line
- Royal Caribbean Cruises Ltd
- OKEE Maritime GmbH
- MOL Chemical Tankers Pte Ltd
- TUIcruises (only one ship was in the area)
- Oldendorff Carriers
- Cargill
- Rudolf Schepers Reederei GmbH
- Rudolf A Oetker A/S & Co KG
- Seabulk Tankers Inc
- Crowley Petroleum Services Inc

To get this level of specificity per vessel, we'll want to use Tylar's tenth degree outputs by MMSI ([bymmsi-10](#)), which so far only contains daily files for the month of 2023-01:

vessel-presence-v1-bymmsi-10-2023-01-01.csv

...

vessel-presence-v1-bymmsi-10-2023-01-31.csv

date	cell_ll_lat	cell_ll_lon	mmsi	hours	dist_travelled_km	avg_speed_knots
2023-01-01	-41	173.2	512005516	0.083	0.823	5.461
2023-01-01	-35.2	100.4	994171250	0.083	0.593	3.994
2023-01-01	-35.2	100.4	994171456	0.083	0.607	3.936
2023-01-01	-34.3	100	994165737	0.083	0.942	6.102
2023-01-01	-33.7	151.2	503131570	0.083	0.49	3.185
2023-01-01	-21.5	-34.7	994208027	0.083		3.291

2023-01-01	-15.2	-30.5	994008823	0.083	0.489	3.165
2023-01-01	-11.8	45.9	620993000	0.083	1.595	10.078
2023-01-01	-10.9	158.4	668000002	0.083	1.421	9.207
2023-01-01	-8.3	94.2	994171030	0.083	0.019	0.123

Will the rest of the year be available? We'll want to summarize by month to compare with the monthly SDMs:

Garrison et al. (2024) [A Density Surface Model Describing the Habitat of the Critically Endangered Rice's Whale Balaenoptera Ricei in the Gulf of Mexico](#). *Endangered Species Research*

Visit [shiny.marinesensitivity.org/sdm](#) to see monthly μ and cv values

As for "Tag-derived depth data", I only know of [Kok et al \(2023\)](#) with strong diel patterns:

Figure1. Dive patterns (black line) of Rice's whales (top) Milky Way and (bottom) Edna. Period between sunset and sunrise shaded gray. Dots depict foraging lunges (red), depth of fluking cessation during ascent (light blue), and descent (dark blue). Blue dashed line designates median neutral buoyancy based on the cessation of fluking during descent and ascent. Note how Milky Way ceased fluking at a shallower depth on the descent vs the ascent (dark blue dots above dashed line, light blue dots below dashed line), while Edna fluked to the same depth on both descent and ascent (light blue and dark blue dots typically at same depth).

Lance Garrison is coauthor or lead author on all of these papers, so hopefully he can be enlisted to advise.

Michael, I'm not sure how to estimate time to produce Deliverables in the [Statement of work. 21Jan25](#) with these latest methods until I have all the data in hand, but presuming all is available I think I could knock it out in a month's time (15 hrs / wk * 4 wks = 60 hrs).

Cheers, Ben

On Wed, Jan 29, 2025 at 1:18 PM Ben Best <ben@ecoquants.com> wrote:

Wow, nice work! Ok, will dig into this recent Science article next week along with others and report back.

On Wed, Jan 29, 2025 at 10:31 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

Doug, Callie, and I helped with a global analysis that was recently published in [Science](#) mapping ship strike risk for blue, fin, humpback, and sperm whales that was led by Anna Nisi from the University of Washington. At the moment we don't have plans to serve this data up on Whale Safe, although certainly using it to help with conversations behind the scenes and we are contemplating trying to host a global workshop later this year to discuss the results and road map mitigation measures in some of these hot spot areas. I did also hear rumblings that IWC might try to host a technical workshop to standardize/update some of the methodology for assessing ship strike risk globally. I can keep you all in the loop on that if we hear anything since it would be relevant for this.

If you have any questions on how Anna calculated ship strike risk for that analysis, happy to connect you.

Rachel

On Tue, Jan 28, 2025 at 2:19 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Yes, all that sounds great. Having those different options will be helpful. If you can provide rough deliverable dates, that would help as well.

I'll get back to you on budget/ caps, as I need to make sure my budget tracker is completely up to date.

I'd defer to Rachel on broader application, of course. Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Tuesday, January 28, 2025 10:25 AM
To: Rachel Rhodes <rachelrhodes@ucsb.edu>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael and Rachel,

This is all fantastic! I need to review all the following methods and papers to compare with the data we have available and goals laid out already in the Statement of Work:

- 1. simple-speed-vessel-risk**
Conn & Silber (2013) [Vessel speed restrictions reduce risk of collision-related mortality for North Atlantic right whales](#). *Ecosphere*
- 2. shipstrike**
Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*
Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*
- 3. whalestrike**
Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*
Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*
- 4. VesselStrikeRiskModel** (SEFSC, Lance Garrison)
Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*

This will take some time to review and I have multiple deadlines for existing projects this week. I can start next week and keep track of my time and lay out different tiers of options for analysis, visualizations and a manuscript. Does that work for you Michael? Feel free to suggest budgets / caps with me.

These methods could also be expanded globally across whale species using GFW data and served via WhaleSafe. I don't want to distract from the species and region of interest here, but want to gauge interest in this broader vision with you all.

Cheers, Ben

On Tue, Jan 28, 2025 at 10:07 AM Rachel Rhodes <rachelrhodes@ucsb.edu> wrote:

Hi Ben and Michael,

I wanted to pass along the sample data that Tyler from Global Fishing Watch created for January 2023 as a sample. Please don't share outside of this thread.

Data link: <https://drive.google.com/drive/folders/1e79Abyi0KEEEFNu7EtFD7E-0aLycFGTu?usp=sharing>

Data are provided as daily csv files with global coverage and gridded at 10th degree resolution. There are two formats:

- **byfleet-10:** Vessel presence grouped by vessel class, flag state, and size class (GT)
- **bymmsi-10:** Vessel presence grouped by individual MMSI

Hoping this format will work for the analysis. Let me know if there is other information that would be helpful to pull in. I am not sure how much flexibility we have, but I can pass it along to Tyler and see. Thanks!

Rachel

On Mon, Jan 27, 2025 at 11:12 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Thanks, Ben! A few initial thoughts.

The risk factors from Conn and Silber (2013) are multipliers from a baseline value, where vessels traveling at 10 knots or below (the baseline) are given a value of 1, and where the multipliers represent the increased strike risk incurred at greater speeds. To my mind, they don't have units per se, or at least non-quantitative me can't think of how to articulate those units.

I guess the larger question is whether you think it's possible to incorporate the new NOAA methodology, at least in part, given the data we can access. It seems that knowing vessel length is very useful and important, though I'm not sure it's essential to using the latest Garrison or Blondin values. Ben, you can gauge how spotty that information is in the attached "individual vessel" table, which Rachel produced (attached). Just scrolling through, I got the sense that the vessel length coverage is quite high—over 90% of entries? And though I haven't checked, I'm guessing that coverage for the 16 major operators we're targeting for outreach will be closer to 100%. Of course, there's the analytical question about what to do with the <10% of unknowns.

Ben, happy to have a call—whatever you think would be most helpful for working this out. Also would be important for me to have a cost/timing estimate, but don't know if you're able to do that at this point.

Thanks!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Thursday, January 23, 2025 2:52 PM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Whoops, while this paper is great:

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

The actual references for this R package include the biophysical model previously mentioned:

· [whalestrike](#)

Kelley et al (2021) [Assessing the lethality of ship strikes on whales using simple biophysical models](#). *Marine Mammal Science*

Kelley et al (2024) [whalestrike: An R package for simulating ship strikes on whales](#). *Journal of Open Source Software*

And now I'm seeing this recent paper with Lance Garrison and a SEFSC risk model with R scripts:

- Blondin et al (2025) [Vessel Strike Encounter Risk Model Informs Mortality Risk for Endangered North Atlantic Right Whales along the United States East Coast](#). *Scientific Reports*
- github.com/SEFSC/VesselStrikeRiskModel

I've placed these and a few other papers here: [literature](#).

On Thu, Jan 23, 2025 at 12:29 PM Ben Best <ben@ecoquants.com> wrote:

Hi Michael,

Thank you so much for drafting the Statement of Work, carrying out all the background research with European partners and laying out an analytical path forward.

Can you explain the **units** of the suggested "risk values" from Conn and Silber (2013):

- (a) the risk factor for vessels traveling at 10 knots or below should have a value of **1**;
- (b) the risk factor for vessels traveling above 10 but at/ below 12 knots should be **1.6** (using the midpoint speed of 11 knots);
- (c) the risk factor for vessels traveling above 12 but at/below 15 knots should be **5.4** (using the midpoint speed of 13.5 knots); and
- (d) the risk factor for vessels traveling above 15 knots should be **10.8** (using the threshold speed of 15 knots)

Ideally, we could translate speed reductions and area avoidance into the number of whales spared, i.e. units of whales saved with error bars.

Applying vessel size of vessel would also be ideal, but this information can be spotty based on ship's MMSI. Rachel would be able to assess availability of this information from Global Fishing Watch and WhaleSafe lookup to their own purchased IMS data. I'm not familiar with Lance's unpublished model, but definitely amenable to using this information to calculate improved estimates. I'm also seeing a biophysical model ([Kelley et al, 2020](#)) and a couple of R packages to facilitate calculation:

- [shipstrike](#)

Keen et al (2022) [A simulation-based tool for predicting whale-vessel encounter rates](#). *Ocean & Coastal Management*

Keen et al (2023) [Ship-strike forecast and mitigation for whales in Gitga'at First Nation territory](#). *ESR*

- [whalestrike](#)

Blondin et al (2020) [Combining high temporal resolution whale distribution and vessel tracking data improves estimates of ship strike risk](#). *Biological Conservation*

Happy to figure this out as we go and/or hop on a call. Look forward to digging into this with you all!

Cheers, Ben

On Tue, Jan 21, 2025 at 3:38 PM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Ben and Rachel – Apologies for the long radio silence. We were occupied with a number of things in the aftermath of the Nov. election that took us pretty far afield. I hope you've both been having a good start to the new year!

We went ahead and drafted a statement of work for the risk-reduction analysis, after getting additional input from our European colleagues.

One question to resolve is which risk values to use in integrating vessel speed. Our colleagues suggested two possible approaches: their own past approach, similar to what they did for shipowner outreach in the Mediterranean, which would apply relative strike-risk values from Conn and Silber (2013) to our existing speed bins; and an approach that incorporates the new, more nuanced values from Garrison et al. (in press), which take account of both vessel speed and vessel length. Using the Garrison et al. values would certainly be preferable—at least for publication—but may be less straightforward. We've suggested making the decision after speaking with Lance, esp. as his work is still in press.

Let us know what you think of the draft statement. Thanks!

Michael

From: Rachel Rhodes <rachelrhodes@ucsb.edu>
Sent: Thursday, November 14, 2024 2:49 PM
To: Ben Best <ben@ecoquants.com>
Cc: Jasny, Michael <mjasny@nrdc.org>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Ben, Michael, and Leyi,

Good news, I just connected with Tyler at Global Fishing Watch and he is working on a prototype of a monthly gridded dataset for ship strike work that should be ready in the next few weeks which we can use. If he gets delayed for some reason, I can always pull our own version of the monthly gridded data set to share with Ben (I just can't share raw AIS data).

Thanks,

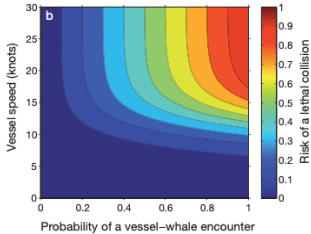
Rachel

On Wed, Nov 6, 2024 at 9:16 AM Ben Best <ben@ecoquants.com> wrote:

Hi Michael, Rachel and Leyi,

Great to chat with you all Monday. I presume we'll wait for further word from Rachel about getting the OK from Global Fishing Watch about proceeding. Then I am happy to help draft a statement of work involving risk analysis, including summarization of the rice's whale SDM to the 7 polygons.

Meanwhile here's a few references I mentioned:

- TEDxCambridge talk [How to motivate yourself to change your behavior](#) by Tali Sharot:
 - Vanderlaan et al (2008) [Reducing the risk of lethal encounters: vessels and right whales in the Bay of Fundy and on the Scotian Shelf](#). *Endangered Species Research*
- 
- Rockwood et al (2017) [High mortality of blue, humpback and fin whales from modeling of vessel collisions on the U.S. West Coast suggests population impacts and insufficient protection](#). *PLOS ONE*

Cheers, Ben

PS Attached is a transcript and AI generated summary, action items from our meeting.

On Mon, Nov 4, 2024 at 11:33 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Looks like we’re on for the new 11am PT/ 2pm ET time. Thanks, both!

Michael

From: Ben Best <ben@ecoquants.com>
Sent: Monday, November 4, 2024 8:16 AM
To: Jasny, Michael <mjasny@nrdc.org>
Cc: Rachel Rhodes <rachelrhodes@ucsb.edu>; Chen, Leyi <lchen@nrdc.org>
Subject: Re: push back one hour?

Hi Michael,

No worries. Works for me.

Thanks, Ben

On Mon, Nov 4, 2024 at 11:00 AM Jasny, Michael <mjasny@nrdc.org> wrote:

Hi Rachel and Ben – Apologies, but I just realized that I have a conflicting meeting on my calendar. It involves a half-dozen people, so isn’t one that I can easily shift. Would it be possible to push back our call by one hour, to 11am PT/ 2pm ET? That time looks free on Ben’s schedule, as does 12pm PT/ 3pm ET.

Please let me know, and apologies again for the conflict!

Michael

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NATURAL RESOURCES DEFENSE COUNCIL

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Please save paper.
Think before printing.

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2/5/26, 8:20 AM

EcoQuants Mail - Further analysis of Rice's whale ship-strike risk

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